

Storage Parameter Estimation: Simplified Poroelasticity Method (Becker 2022)

From General to Simplified

Wang (2000) Eq. 4.65

$$\alpha \frac{\partial \epsilon}{\partial t} + S_{\epsilon} \frac{\partial p}{\partial t} = \frac{k}{\mu} \nabla^2 p + Q$$

Observation well: $\nabla^2 p \approx 0, Q = 0$

Simplified Equation (Solved for S_{ϵ})

$$S_{\epsilon} = \frac{\alpha}{\gamma} \cdot \frac{\partial \epsilon / \partial t}{\partial h / \partial t}$$

Biot-Willis coefficient

Strain rate (DAS)

Specific weight of water

Drawdown rate (piezometer)

Required Measurements

(Recovery Period Only)

Strain Rate

1

$$\dot{\epsilon}_{zz}(z, t)$$

Distributed Acoustic Sensing (DAS)

ROI across pumping interval

Drawdown Rate

2

$$\dot{h}(t)$$

Paired PM-07
Piezometer Zone

Selected by response time (β)

Analysis Workflow

① EXTRACT RECOVERY DATA

DAS: $\dot{\epsilon}(t)$ across ROI | PM-07: $\dot{h}(t)$

- 75-second recovery window
- Simultaneous measurements

② LINEAR REGRESSION

$$\text{slope} = \frac{\dot{\epsilon}}{\dot{h}} = \frac{\partial \epsilon / \partial t}{\partial h / \partial t}$$

- Strain: single-pass moving average (20s)
- Drawdown: Bourdet derivative + 15s smoothing

③ CALCULATE STORAGE

$$S_{\epsilon} = \frac{\alpha}{\gamma} \cdot \text{slope} \rightarrow S_s \approx S_{\epsilon} \gamma$$

- $\alpha = 1.0$ (unconsolidated alluvium)
- $\gamma = 9810 \text{ N/m}^3$
- slope from regression

★ Result: Direct Poroelastic Storage from Slope of Strain vs. Drawdown! ★